

3.

BARS MARKED CONTINUOUS TO BE TERMINATED IN STANDARD HOOKS AT ENDS AND SPLICED WITH ADJACENT BARS USING CLASS B LAPS. FOR LAP LENGTHS AND DEVELOPMENT LENGTHS, REFER TO TYPICAL DETAILS TC-REINF-01.
4.

ALL REBAR HOOKS TO BE STANDARD LENGTH 90° OR 180° HOOKS. REBAR LENGTHS LISTED ON DRAWINGS DO NOT INCLUDE THE HOOK LENGTH.
5.

UNLESS A SPECIFIC STIRRUP SHAPE IS INDICATED ON PLANS OR SCHEDULES, ALL STIRRUPS TO BE CLOSED HOOPS. NUMBER OF STIRRUPS DENOTES THE NUMBER OF FULL STIRRUPS, EACH HAVING TWO LEGS.
6.

WHERE TWO BARS OF DIFFERENT SIZE ARE LAPPED IN TENSION, SPLICE LENGTH TO BE EQUAL TO THE SMALLER BAR'S TENSION LAP SPLICE, OR TO THE LARGER BAR'S TENSION DEVELOPMENT LENGTH, WHICHEVER IS LONGER.
7.

WHERE TWO BARS OF DIFFERENT SIZE ARE LAPPED IN COMPRESSION, SPLICE LENGTH TO BE EQUAL TO THE SMALLER BAR'S COMPRESSION LAP SPLICE, OR TO THE LARGER BAR'S COMPRESSION DEVELOPMENT LENGTH, WHICHEVER IS LONGER.
8.

FOR BUNDLED BARS, FULLY STAGGER SPLICES OF EACH INDIVIDUAL BAR IN THE BUNDLE.
9.

LAP WELDED WIRE FABRIC SHEETS BY ONE SPACING OF CROSS WIRES + 50 (2"), MEASURED BETWEEN THE OUTERMOST CROSS WIRES IN EACH SHEET.
10.

PROVIDE ADDITIONAL SUPPORT BARS AS REQUIRED TO ADEQUATELY SUPPORT AND SECURE ALL REINFORCEMENT AND PREVENT MOVEMENT WHEN PLACING CONCRETE.
11.

PROVIDE SUFFICIENT CHAIRS TO REINFORCING TO MAINTAIN SPECIFIED CONCRETE COVER.
12.

PLACE WELDED WIRE FABRIC IN SLABS ON GRADE AT 1/3 SLAB THICKNESS BELOW TOP OF SLAB. PROVIDE ADEQUATE CHAIRS TO KEEP IN SPECIFIED POSITION. LIFTING WWF AFTER CONCRETE IS POURED TO BRING IT IN POSITION IS NOT ACCEPTABLE.
13.

ALL REINFORCING TO BE CLEAN, FREE OF LOOSE SCALE, OIL, DIRT, RUST, AND ANY OTHER FOREIGN COATING THAT AFFECT BONDING CAPACITY.
14.

MINIMUM CLEAR SPACING BETWEEN ADJACENT BARS TO BE AT LEAST 1.4 TIMES THE BAR DIAMETER OR 1.4 TIMES THE NOMINAL MAXIMUM SIZE OF THE COARSE AGGREGATE, WHICHEVER IS MORE.
15.

WHERE PARALLEL REINFORCEMENT IS PLACED IN TWO OR MORE LAYERS, POSITION BARS IN UPPER LAYER DIRECTLY ABOVE THE BARS IN LOWER LAYER, MAINTAINING THE MINIMUM CLEAR SPACING BETWEEN LAYERS AS SPECIFIED ABOVE.
16.

UNLESS NOTED OTHERWISE ON DRAWINGS, MINIMUM CONCRETE COVER TO PRINCIPAL REINFORCEMENT (EXCLUDING WWF) TO BE AS FOLLOWS (ALSO REFER TO TYPICAL DETAIL TC-REINF-21):

18.

WALLS
1.

UNLESS OTHERWISE NOTED ON DRAWINGS, SCHEDULES OR NOTES, MINIMUM REINFORCEMENT FOR CONCRETE WALLS TO BE AS FOLLOWS:
150 (6") MAXIMUM WALL: 10M @ 330 (13") H + 10M @ 450 (18") V IN CENTRE
200 (8") MAXIMUM WALL: 10M @ 250 (10") H + 10M @ 330 (13") V IN CENTRE
250 (10") MAXIMUM WALL: 10M @ 400 (16") HEF + 10M @ 500 (20") VEF
300 (12") MAXIMUM WALL: 10M @ 330 (13") HEF + 10M @ 440 (17") VEF
VERTICAL REINFORCING TO BE OUTSIDE LAYER. ALL SPLICES TO BE CLASS "B".
2.

REFER TO TYPICAL DETAIL TC-WALL-01 FOR REINFORCING REQUIRED AT WALL CORNERS AND INTERSECTIONS.
3.

REFER TO TYPICAL DETAIL TC-WALL-11 FOR MINIMUM DOWELING REQUIREMENTS BETWEEN WALLS AND SLABS.
4.

FOR OPENINGS LARGER THAN 400 x 400 (16"x16"), PROVIDE ADDITIONAL REINFORCING AS SHOWN ON TYPICAL DETAIL TC-WALL-32.
5.

FOR SHEAR WALL REQUIREMENTS, REFER TO SHEAR WALL NOTES ON DRAWINGS S ____.
6.

FOR MASONRY WALL REINFORCEMENT REFER TO MASONRY NOTES.
19.

SLABS
9.

UNLESS OTHERWISE SPECIFIED ON PLANS, PROVIDE BOTTOM TEMPERATURE REINFORCEMENT IN BOTH DIRECTIONS FOR FRAMED SLABS IN ACCORDANCE WITH TYPICAL DETAIL TC-SL-02.
10.

UNLESS NOTED OTHERWISE, DO NOT ELIMINATE OR CUT REINFORCEMENT TO ACCOMMODATE MECHANICAL AND ELECTRICAL SLEEVES, OPENINGS OR HARDWARE. SPREAD REINFORCING AROUND SLEEVES.
11.

TOP BARS ARE SHOWN AS: _____
BOTTOM BARS ARE SHOWN AS: _ _ _ _ _
12.

FOR OPENINGS LARGER THAN 400 x 400 (16"x16"), PROVIDE ADDITIONAL REINFORCING AS SHOWN ON TYPICAL DETAIL TC-SL-62.
13.

WHERE IN FLOOR HEATING IS CAST INTO SLABS, PROVIDE ADDITIONAL LAYER OF WWF 102 X 102 MW18.7 X MW18.7 WHERE REQUIRED FOR IN FLOOR HEATING SECUREMENT.

POST-INSTALLED ANCHORS AND DOWELS

1.

UNLESS OTHERWISE NOTED, PROVIDE STRUCTURAL ANCHORS AS FOLLOWS:
1.

WHERE DRILLED CONCRETE ANCHORS (DCA) OR DRILLED MASONRY ANCHORS (DMA) ARE NOTED ON DRAWINGS, PROVIDE HILTI KWIK BOLT — T22 EXPANSION ANCHORS. TIGHTEN USING THE TORQUE SPECIFIED BY THE MANUFACTURER. LOCATE DMA MIN. 35 (1-3/8") FROM ANY VERTICAL MORTAR JOINT. DO NOT INSTALL DMA INTO HOLLOW MASONRY, ADVISE WSP-S IF HOLLOW MASONRY IS FOUND ON SITE WHERE DMA HAVE BEEN SPECIFIED.
2.

WHERE ADHESIVE CONCRETE ANCHORS (ACA) ARE NOTED ON DRAWINGS, PROVIDE HILTI HIT-HY 200 ADHESIVE ANCHORING SYSTEM WITH HILTI HIT-Z ANCHOR RODS
3.

WHERE ADHESIVE MASONRY ANCHORS (AMA) ARE NOTED ON DRAWINGS, PROVIDE HILTI HIT-HY 270 ADHESIVE ANCHORING SYSTEM WITH HAS-V THREADED RODS.
4.

WHERE REBAR DOWEL ANCHORS (RDA) ARE NOTED ON DRAWINGS, PROVIDE HILTI HIT-RE 500 V3 (FOR ANCHORAGE TO CONCRETE) OR HILTI HIT-HY 270 (FOR ANCHORAGE TO SOLID OR GROUTED MASONRY) ADHESIVE ANCHORING SYSTEM INSTALLED USING HILTI SAFESIT HOLLOW DRILL BIT TECHNOLOGY
5.

WHERE HOLLOW MASONRY ANCHORS (HMA) ARE NOTED ON DRAWINGS, PROVIDE HILTI HIT—HY 270 ADHESIVE ANCHORING SYSTEM WITH HIT-SC MESH SLEEVE AND HAS-V THREADED RODS OR APPROVED EQUIVALENT.
2.

ANCHORS LOCATED OUTSIDE THE BUILDING ENVELOPE'S VAPOUR BARRIER TO BE HOT DIP GALVANIZED.
3.

CONCRETE TO BE MINIMUM 28 DAYS OLD AT THE TIME OF ANCHOR INSTALLATION.
4.

USE DRILLING AND INSTALLATION TOOLS AND PROCEDURES PER MANUFACTURER'S RECOMMENDATIONS. DO NOT CORE DRILL UNLESS SPECIFICALLY NOTED ON DRAWINGS. HOLE DIAMETERS NOT TO EXCEED THOSE REQUIRED BY MANUFACTURER.
5.

WHERE CORE DRILLING IS SPECIFIED, CLEAN AND ROUGHEN HOLES PER MANUFACTURER'S RECOMMENDATION.
6.

ARRANGE FOR THE ANCHOR MANUFACTURER TO CONDUCT ON SITE TRAINING FOR INSTALLATION OF ALL THE PRODUCTS SPECIFIED, AND FOR ALL CONDITIONS ENCOUNTERED (E.G. HORIZONTAL, INCLINED, OVERHEAD). ALL INSTALLERS MUST COMPLETE THE SUPPLIER CERTIFIED INSTALLER TRAINING PROGRAM. SUBMIT COPIES OF COMPLETION CERTIFICATES FOR WSP-S RECORD.
7.

ARRANGE FOR A MANUFACTURER'S TECHNICAL REPRESENTATIVE TO BE PRESENT DURING INSTALLATION OF FIRST FEW ANCHORS OF EACH SIZE AND TYPE. SUBMIT SITE REPORTS INDICATING ANCHOR TYPES AND SIZES INSTALLED, LOCATIONS AND INSTALLERS' NAMES.
8.

ANCHOR AND DOWEL CAPACITY IS DEPENDENT UPON SPACING BETWEEN ADJACENT ANCHORS AND THEIR PROXIMITY TO CONCRETE AND MASONRY EDGES; THEREFORE, ALL ANCHORS MUST BE INSTALLED WITH CLEARANCES AND EDGE DISTANCES INDICATED ON DRAWINGS
9.

UNLESS CORE DRILLING IS SPECIFIED ON DRAWINGS, DO NOT CUT REINFORCEMENT TO ACCOMMODATE DRILLED ANCHORS AND DOWELS. SCAN THE STRUCTURE TO LOCATE REINFORCEMENT PRIOR TO FABRICATING STRUCTURAL STEEL FASTENED BY DRILLED ANCHORS.
10.

WHEN OBSTRUCTIONS PREVENT DRILLING HOLES IN SPECIFIED LOCATIONS TO THE REQUIRED DEPTH, RELOCATE AT NO EXTRA COST TO THE CONTRACT. OBTAIN WSP-S APPROVAL OF NEW LOCATIONS BEFORE DRILLING. MODIFICATIONS TO CONNECTED MEMBERS AND ADDITIONAL ANCHORS / DOWELS MAY BE REQUIRED. FILL ABANDONED HOLES WHICH ARE CLOSER THAN 3 TIMES THE HOLE DIAMETER FROM THE RELOCATED ANCHORS WITH HILTI HY-200 ADHESIVE OR WITH 30 MPa GROUT. DO NOT TIGHTEN ANCHORS UNTIL THE FILLER HAS FULLY CURED.
11.

UNLESS OTHERWISE NOTED ON DRAWINGS, EMBEDMENT LENGTHS FOR POST-INSTALLED HILTI ANCHORS TO BE:

ANCHOR SIZE	EXPANSION ANCHORS		ADHESIVE ANCHORS			
	INTO CONCRETE (DCA)	INTO MASONRY (DMA)	INTO CONCRETE (ACA)	INTO SOLID OR GROUTED MASONRY (AMA)	INTO HOLLOW MASONRY (HMA)	
	KB-TZ2		HIT-Z ROD	HAS-V ROD	HAS-V ROD + HIT-SC SCREEN	
			HIT HY-200 ADHESIVE	HIT HY-270 ADHESIVE	HIT HY-270 ADHESIVE	
10 (3/8")	64 (2-1/2")		86 (3-3/8")		50 (2")	80 (3-1/8")
12 (1/2")	83 (3-1/4")		114 (4-1/2")		50 (2")	80 (3-1/8")
16 (5/8")	102 (4")		143 (5-5/8")		-	-
19 (3/4")	121 (4-3/4")		171 (6-3/4")		-	-
25 (1")	146 (5-3/4")		—		-	-

NOTES:

1.

ALL EMBEDMENT LENGTHS SHOWN ARE EFFECTIVE EMBEDMENT LENGTHS; FOR REQUIRED HOLE DEPTHS FOLLOW HILTI RECOMMENDATIONS.
2.

SEE DRAWINGS FOR EMBEDMENT LENGTHS OF REBAR DOWEL ANCHORS (RDA).
12.

DO NOT BEND POST INSTALLED DOWELS AND RODS AFTER INSTALLATION.
13.

DO NOT WELD TO PLATES FASTENED WITH ADHESIVE ANCHORS AFTER THE ADHESIVE IS PLACED.

STRUCTURAL STEEL

1.

CONFORM TO CSA S16.
2.

MATERIALS: TO CSA G40.21 UNLESS OTHERWISE NOTED, WITH THE FOLLOWING GRADES:

W, WWF AND S SECTIONS, CHANNELS AND ANGLES: 350W

PLATES, BARS: 300W

HOLLOW STRUCTURAL SECTIONS: 350W (CLASS "C" OR "H"),

BOLTS: ASTM F3125 GRADE A325, UNLESS NOTED

STAINLESS STEEL BOLTS: ASTM F593HEADED STUDS: CSA W59, TYPE B, min. Fy= 350 MPa

ANCHOR RODS: ASTM F1554 GRADE 36
3.

DETAILS ON STRUCTURAL DRAWINGS SHOW DESIGN INTENT. REFER TO SPECIFICATIONS FOR CONNECTION DESIGN, DETAILING, FABRICATION, AND ERECTION REQUIREMENTS.
4.

CONNECT BEAMS FOR THE FORCES SHOWN; IF NO FORCE IS INDICATED, CONNECT NON COMPOSITE BEAMS FOR THE REACTION DUE TO MAXIMUM UNIFORMLY DISTRIBUTED LOAD CAPACITY OF THE BEAM IN BENDING, AND CONNECT COMPOSITE BEAMS FOR ONE AND A HALF TIMES THE REACTION DUE TO MAXIMUM UNIFORMLY DISTRIBUTED LOAD CAPACITY OF THE NON COMPOSITE SECTION IN BENDING.
5.

UNLESS LSH (LONG SIDE HORIZONTAL) IS INDICATED ON DRAWINGS, PLACE HORIZONTAL TUBULAR MEMBERS WITH LONG SIDE IN THE VERTICAL PLANE.
6.

SPACE BEAMS AND JOISTS EVENLY WITHIN BAYS UNLESS OTHERWISE NOTED ON DRAWINGS.
7.

DO NOT CUT HOLES OR OTHERWISE MODIFY STRUCTURAL MEMBERS ON SITE.
8.

CLEAN SURFACES DOWN TO BARE METAL AND APPLY TWO COATS OF ZINC-RICH TOUCH-UP PAINT TO ANY GALVANIZED SURFACE THAT HAS BEEN DAMAGED OR FIELD WELDED.
9.

PROVIDE ALL ERECTION BRACING REQUIRED TO KEEP THE STRUCTURE STABLE AND IN ALIGNMENT DURING CONSTRUCTION.
10.

PROVIDE 40 MPa NON SHRINK GROUT UNDER BASE PLATES. DO NOT APPLY ANY LOADS TO THE STEELWORK BEFORE GROUT ACHIEVES SUFFICIENT STRENGTH.
11.

WHERE STRUCTURAL DRAWINGS INDICATE PROTECTED ZONES, DO NOT FASTEN OR OTHERWISE CONNECT ANY STRUCTURAL OR NON-STRUCTURAL ELEMENT TO THOSE ZONES (INCLUDING STUDWALL RAILS, CLIPS FOR LATERAL SUPPORT OF MASONRY, HEADED STUDS, ETC), WITH EXCEPTION OF STEEL DECKS CONNECTED WITH 19 (3/4") PUDDLE WELDS. FAILURE TO ADHERE WILL REQUIRE COMPLETE REPLACEMENT OF THE AFFECTED MEMBER.
12.

DISTRIBUTE HANGER LOADS FROM MECHANICAL AND HEAVY ELECTRICAL SERVICES SUSPENDED FROM STEELWORK UNIFORMLY ALONG MEMBERS. ALTERNATE HANGER POSITION ON EITHER SIDE OF MEMBERS.
13.

CONNECT HANGERS FOR MECHANICAL AND ELECTRICAL SERVICES AND OTHER NON STRUCTURAL ELEMENTS AS WELL AS ELEMENTS REQUIRED FOR THEIR SEISMIC RESTRAINT NOT TO CAUSE TWISTING OF STEEL MEMBERS OR EXCESSIVE BENDING OF MEMBER FLANGES.
14.

DO NOT APPLY LATERAL LOADS TO MEMBERS UNLESS APPROVED BY THE CONSULTANT.
2.

AESS MEMBERS: ERECT USING SOFTENED SLINGS OR OTHER METHODS SUCH THAT THEY ARE NOT DAMAGED. PROVIDE PADDING AS REQUIRED TO PROTECT WHILE RIGGING AND ALIGNING. WELD TABS FOR TEMPORARY BRACINGS AND SAFETY CABLING ONLY AT POINTS CONCEALED FROM VIEW IN THE COMPLETE STRUCTURE OR WHERE APPROVED BY WSP-S.

STEEL JOISTS

1.

CONFORM TO CSA S16 AND CSA S136.
2.

MATERIALS: TO CSA G40.21, ASTM A1085 OR ASTM A500.
3.

DESIGN STEEL JOISTS FOR THE LOADS AND STIFFNESSES DESCRIBED ON THE STRUCTURAL DRAWINGS. IN ADDITION, DESIGN TOP AND BOTTOM JOIST CHORDS FOR 1.5 kN POINT LOAD APPLIED ANYWHERE ALONG THEIR LENGTH (UNLESS HIGHER LOAD IS INDICATED ON DRAWINGS). SEE PLANS FOR OTHER POINT LOADS, WALL LOADS, AND MECHANICAL LOADS. SEE TYPICAL DETAIL TS-MISC-02 FOR LOADS FROM ROOF ANCHORS.
4.

DO NOT MODIFY STEEL JOISTS ON SITE.
5.

TERMINATE BRIDGING BY ANCHORING TO FRAMING MEMBERS AND SUPPORTS, OR BY CROSS BRACED BRIDGING BAYS.
6.

PROVIDE TIE JOISTS "TJ" AT COLUMNS FOR JOIST ERECTION STABILITY. UNLESS NOTED OTHERWISE, BOTTOM CHORD ATTACHMENT MUST NOT INDUCE AXIAL LOAD IN JOIST.
7.

WHEN SUSPENDING VERTICAL LOADS FROM TOP OR BOTTOM JOIST CHORDS, DISTRIBUTE HANGERS UNIFORMLY ALONG JOISTS. DO NOT CAUSE TWISTING OF JOISTS OR JOIST CHORDS. EXTEND HANGER RODS BETWEEN DOUBLE ANGLE CHORDS WHERE POSSIBLE; OTHERWISE ATTACH USING ONLY CLAMPS OR U-BOLTS. IF THE APPLIED LOAD IS MORE THAN 1.5 kN, LOCATE SUSPENSION POINT NOT MORE THAN 100 (4") FROM JOIST PANEL POINT.
8.

WHERE SEISMIC RESTRAINTS ARE REQUIRED FOR NON-STRUCTURAL ELEMENTS, APPLY LOADS IN THE PLANE OF THE JOISTS, AND DO NOT EXCEED 1.5 kN. DO NOT APPLY CONCENTRATED MOMENTS.
9.

DO NOT APPLY LATERAL LOADS TO ANY JOISTS.

SHAPE

Architecture Inc.

1462 WEST 7TH AVENUE
VANCOUVER, BC CANADA
V6H 1C1

TEL: (604) 687-4457
EMAIL: info@shapearchitecture.ca
WEB: www.shapearchitecture.ca

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NOTES	
NO	DESCRIPTION

REVISIONS		
NO	DESCRIPTION	DATE
1	ISSUED FOR CLASS B COSTING	2022-07-26
2	ISSUED FOR REVIEW	2022-10-28

CHUCK BAILEY RECREATION CENTRE EXPANSION - PHASE 2

13458 107A Avenue, Surrey, BC V3T 0G4

PRELIMINARY NOT FOR CONSTRUCTION

GENERAL NOTES

DATE	26-07-2022
DRAWN BY	HK
CHECKED BY	CKS
SCALE	
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